

CERN Supervision, Control and Data Acquisition System for Radiation and Environmental Protection

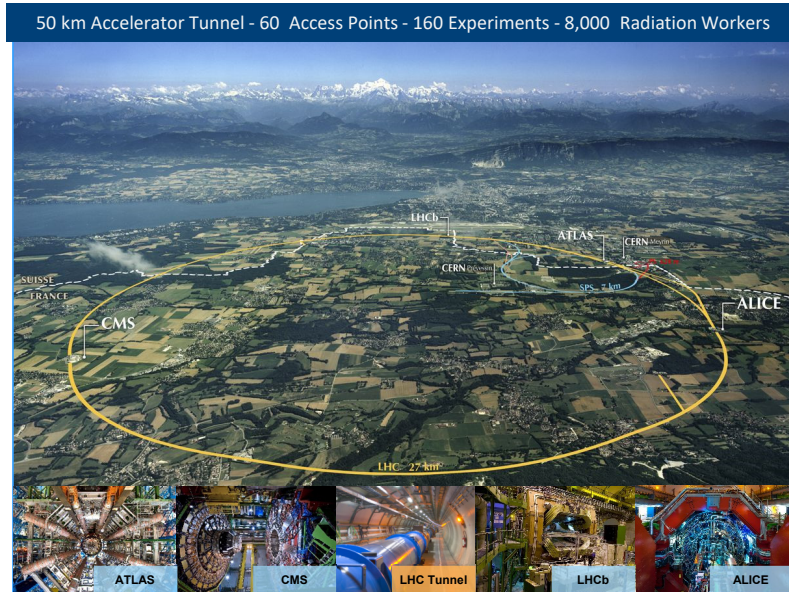
G. SEGURA on behalf of REMUS Team
CERN Health, Safety and Environment Unit



Summary

- Radiation Protection and Environmental monitoring at CERN
- REMUS System
 - Main Functionality
 - System architecture
 - Heterogeneous equipment integration
 - Tailored user interfaces
 - Continuous operation
 - Future evolutions

Radiation Protection and Environment Monitoring at CERN



Radiation Protection and Environment Monitoring at CERN

Environment Monitoring:

50 km Accelerator Tunnel - 60 Access Points - 160 Experiments - 8,000 Radiation Workers



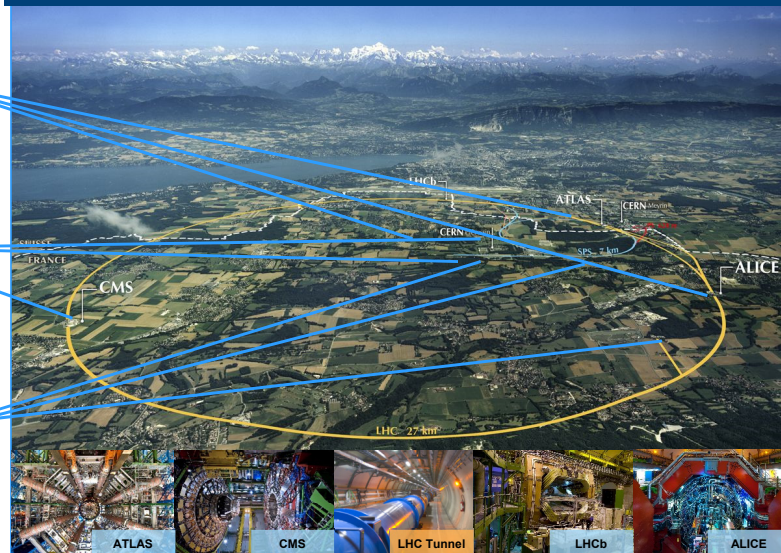
Stray Radiation (x43)



Ventilation (x47)



Water (x14)



Many more!

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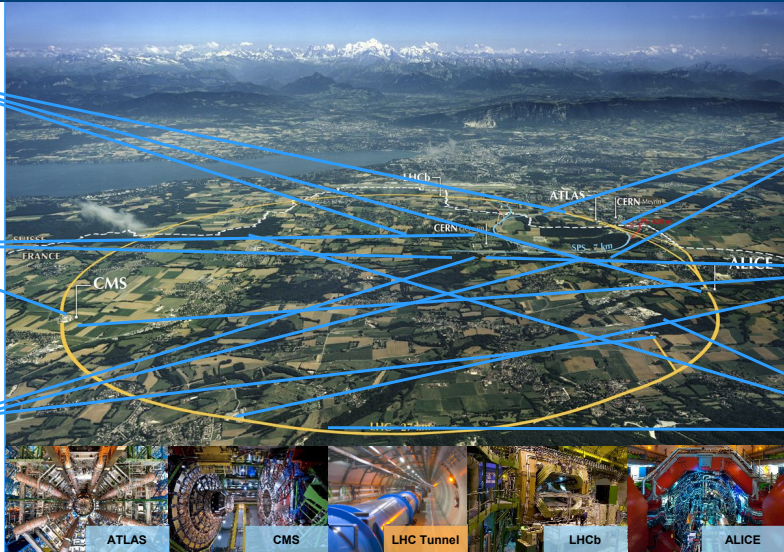


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Radiation Protection:



Gamma Radiation (x46)



Hand-Foot Contamination (x60)



Radiation with Alarm Units (x100)
and interlocks

Many more!

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Many more!

- ~500 Monitoring Stations:
- 24 different types
 - Commercial Off-the-Shelf & Internally Developed Stations
 - Surface and Underground areas

- ~3,200 Measurement Channels:
- 1,000 measurements archived / second

Workplace and Environment Safety

Reporting to authorities:

- Nature and quantities of emitted ionizing radiation
- Conventional environmental measured values



Radiation Protection:



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Hand-Foot Contamination (x60)



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Problem Statement / Role of HSE Supervision

- **What do we need?**
 - SCADA system: Supervision, Control And Data Acquisition system for a large number of devices

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- **What do we need?**
 - SCADA system: Supervision, Control And Data Acquisition system for a large number of devices
- **But much more:**

Distributed	to work with distributed devices
Flexible	to interface heterogeneous devices (vendors, protocols, networks, etc.)
Scalable	to absorb the growing number of instruments
Available	24/7, 365 days a year: Environmental protection does not have technical stops
Multiplatform	Users can use Windows, Linux or mobile devices (read only)
Tailored UI	to be suitable for accelerators and experiment operators, physicists, RP engineers, Environmental engineers, fire brigade, instrumentation maintenance teams, software maintenance team

REMUS - Radiation and Environment Monitoring Unified Supervision

- **REMUS Project**

- Previous HSE supervisory system, **RAMSES**, was unsuitable for long term use:
 - Limited evolutivity
 - Limited scalability
 - Only 9 types of Monitoring Station covered (rest was standalone or used vendor-specific supervisions)

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■ REMUS Project

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- **REMUS** Project Started in 2012, tackling those limitations
 - **Unify** all CERN Radiation and Environment supervisory systems
 - **Reduce delay** and the cost of **adding new devices** to the supervision
 - Provide **light and fast clients, customized** for each user's requirements
 - **Reduce overall maintenance** needs for operation
 - Use CERN **common software & framework** for SCADA:
 - Siemens **WinCC OA**
 - CERN **JCOP**(Joint COntrols Project) framework



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■ Guideline

- **Reliability, Scalability, Performance, Adaptability, Cost-Effectiveness**

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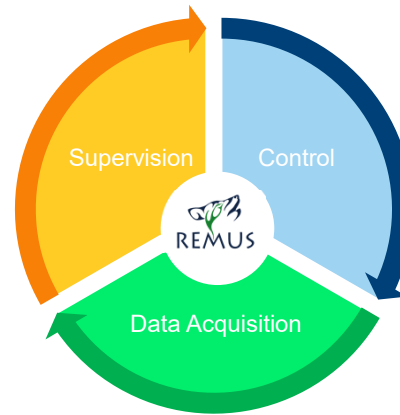
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■ Project Milestones

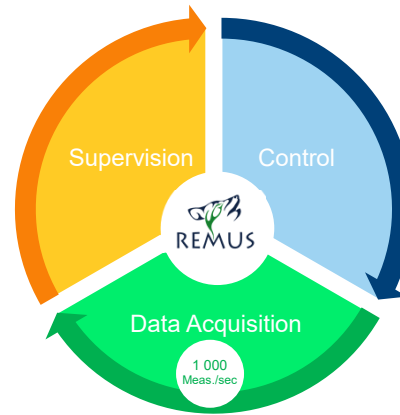
- 2012: Project kick-off
- 2013: First usage of REMUS as a production system
- 2017: Previous supervision (RAMSES) 100% decommissioned
- 2018: REMUS covers 24 types of MS and archives 38 billion measurement a year (RAMSES was 15 billion)

REMUS – Main Functionality



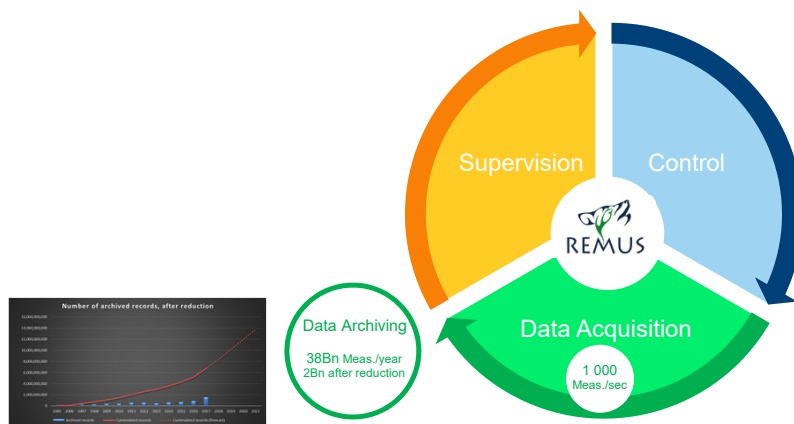
REMUS Functionality

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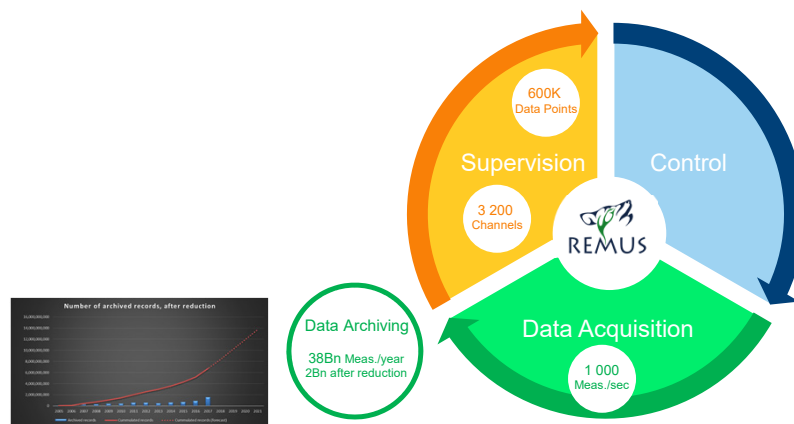
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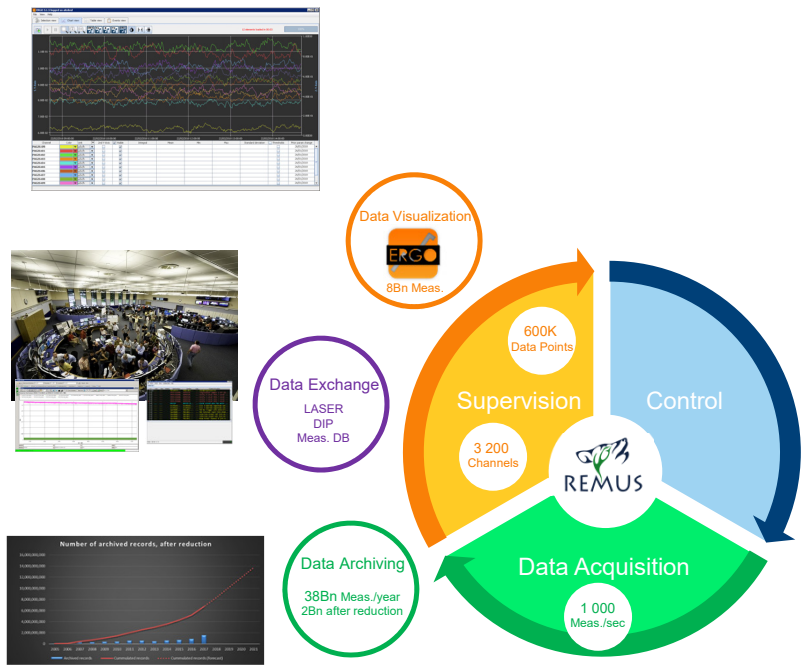
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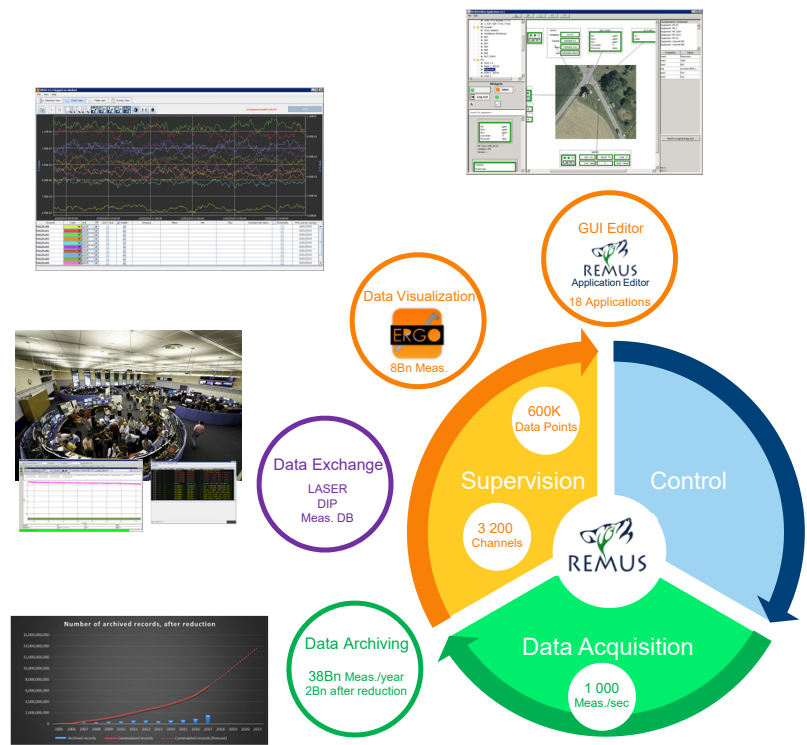
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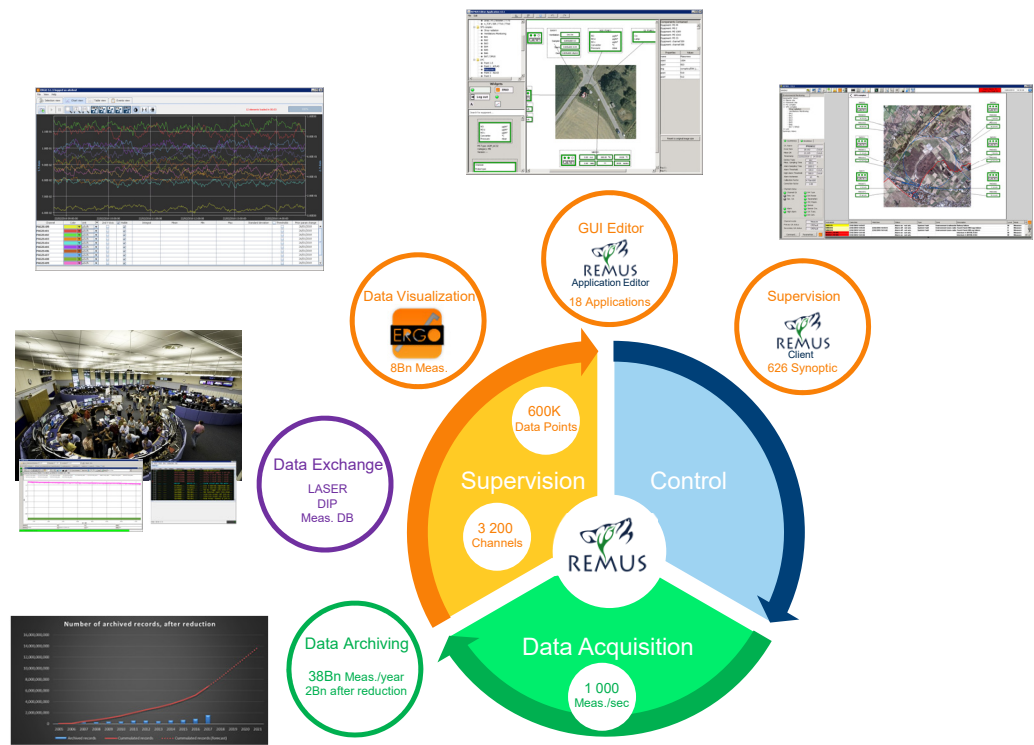


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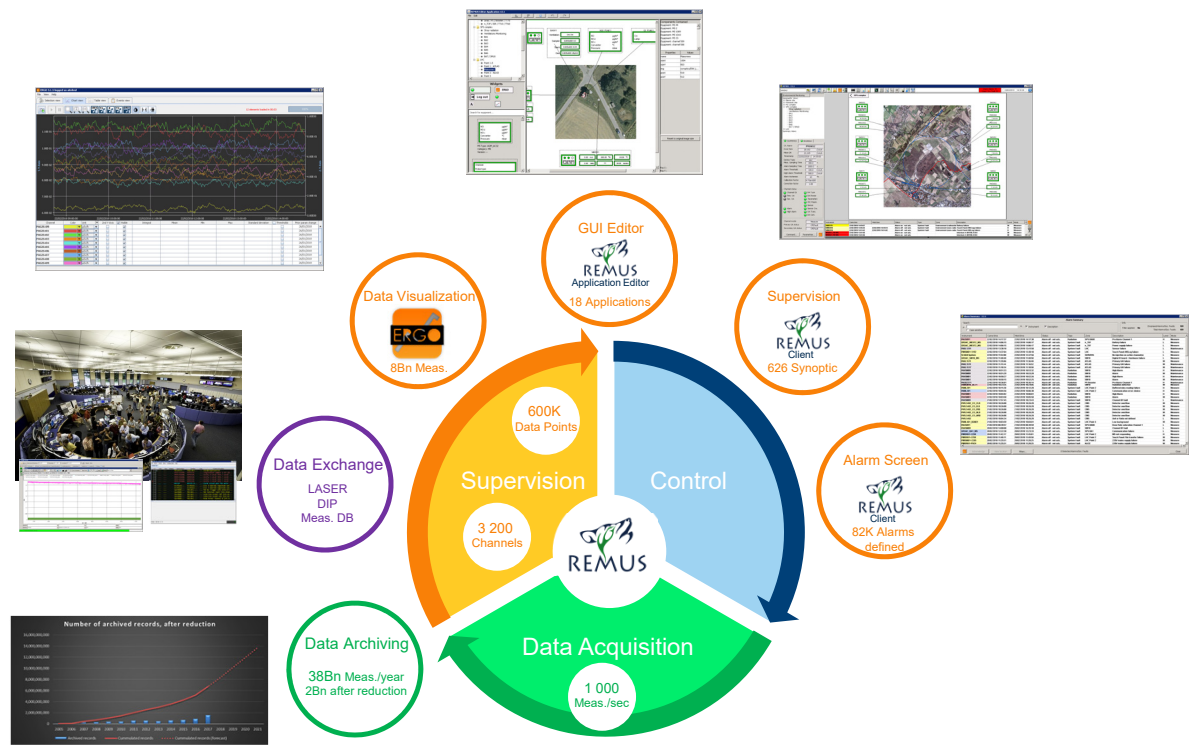


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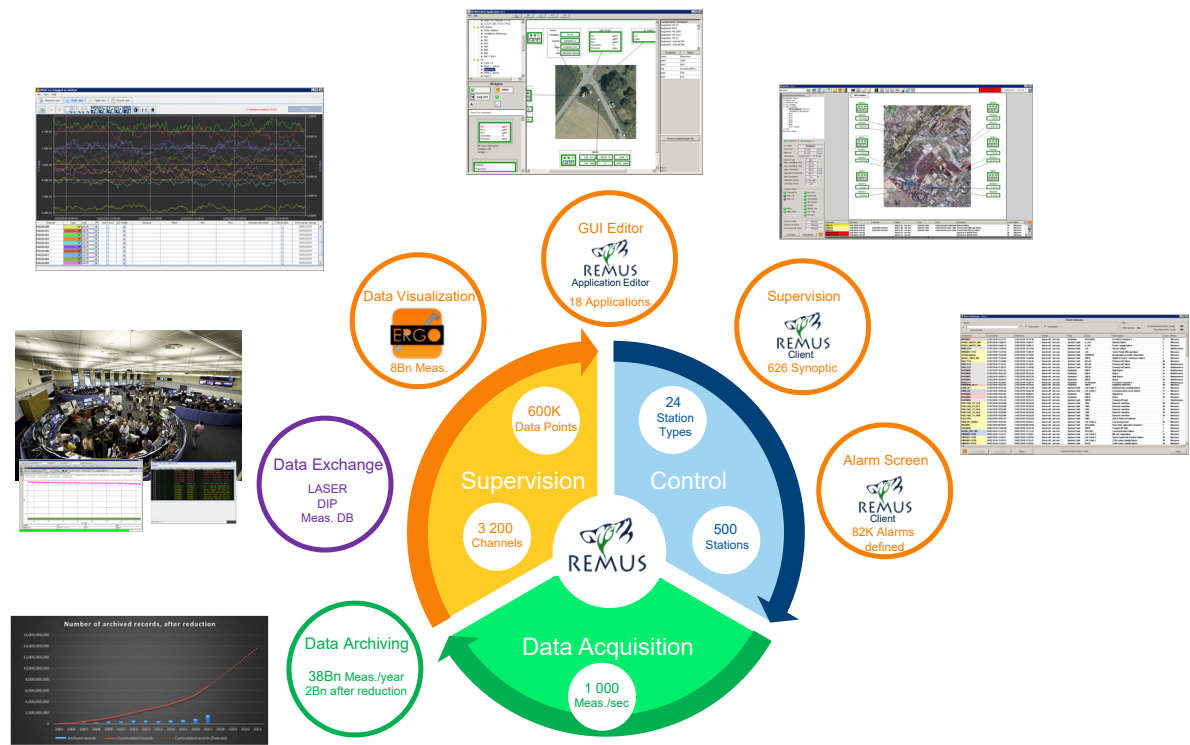


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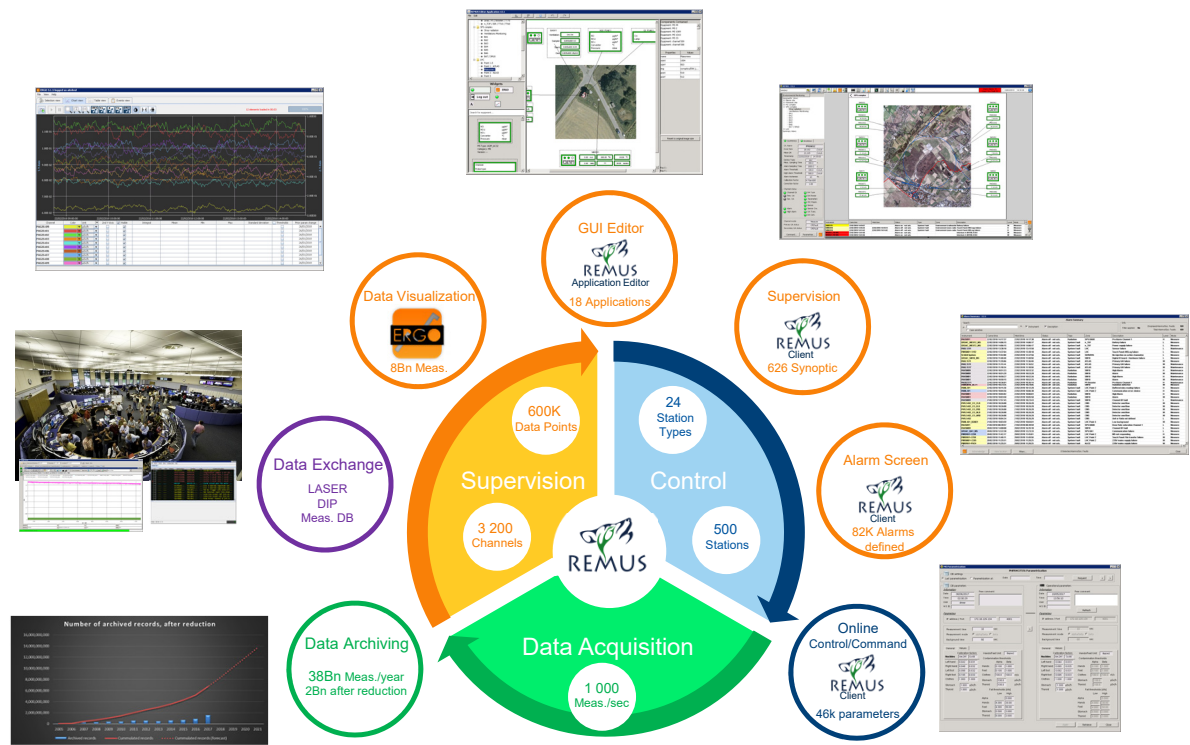


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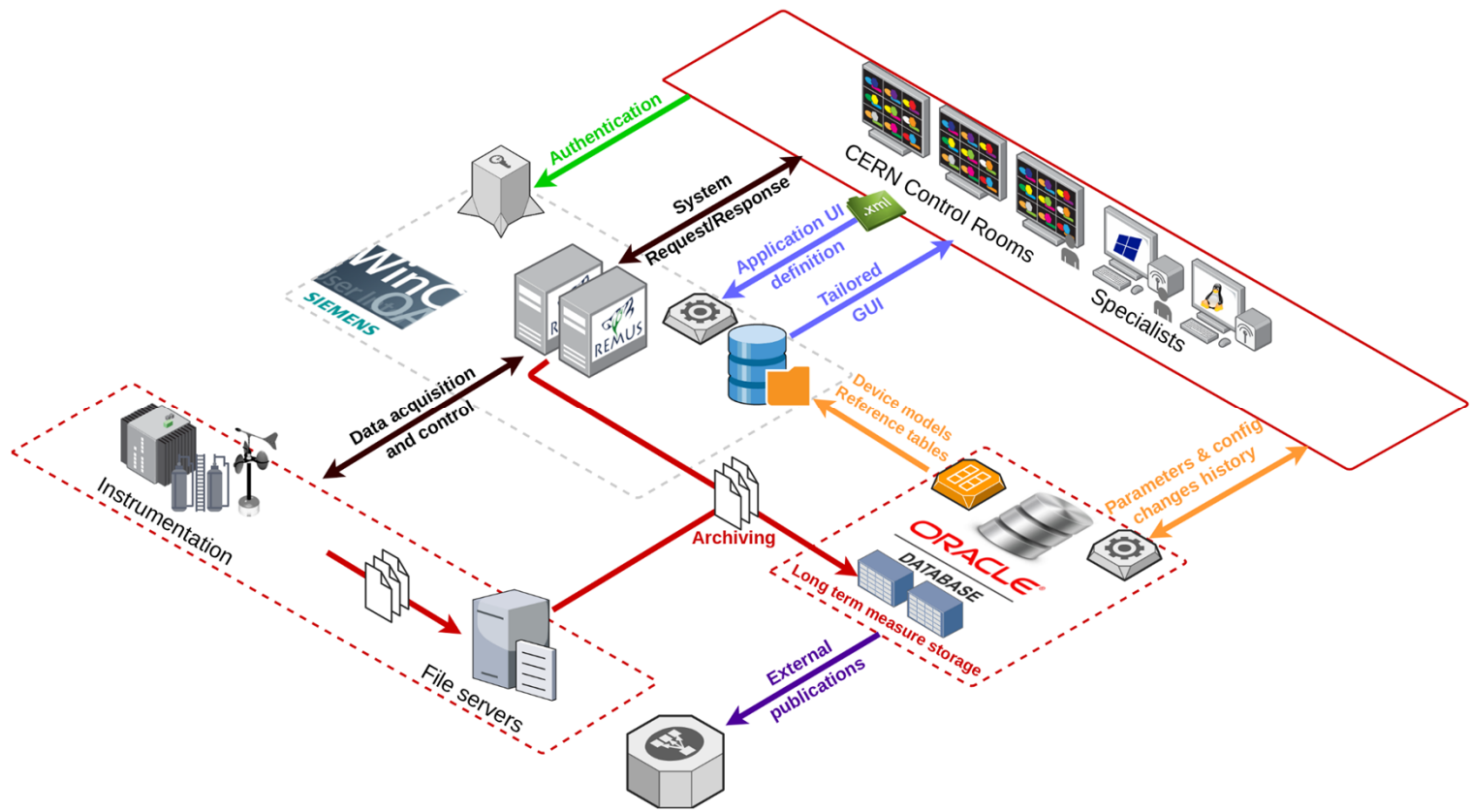


REMUS Functionality

REMUS – Main Functionality



REMUS Architecture



REMUS - Heterogeneous equipment integration handling

■ Challenges

- HSE owns very diverse types of monitoring equipment, for 2 reasons:
 - Wide scope of measurements to acquire
 - Long lifetime of equipment, costly to replace: electronics of different generations overlap
- **This is a common challenge for supervisory systems**

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▪ REMUS Solution

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 - 3 concepts only: Channel, Device, Monitoring Stations
 - All type of instrumentation are modeled using the same concepts

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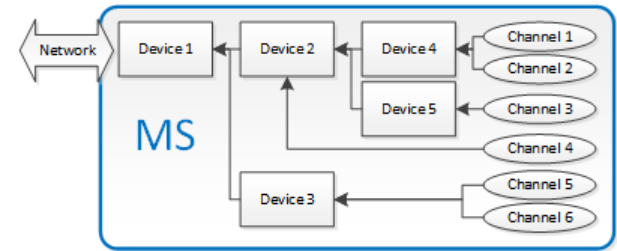
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REMUS Instrumentation Model

Simple to comprehend!

REMUS - Tailored User Interfaces

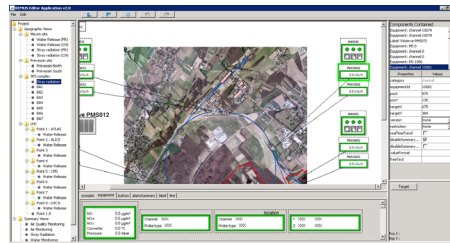
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 - “Drag an drop” User Interface Editor
 - No programming involved in GUIs creation
 - The full process takes few minutes

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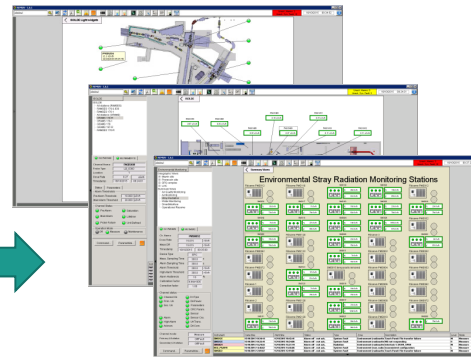
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 - Creator of the GUI determines its access control
 - (visualization, acknowledgment, parametrization, configuration, admin.)
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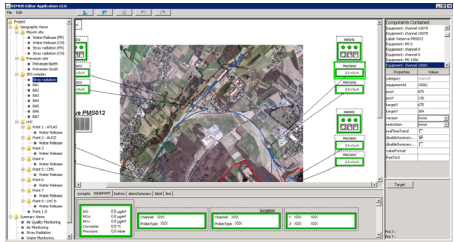
User Interfaces Editor



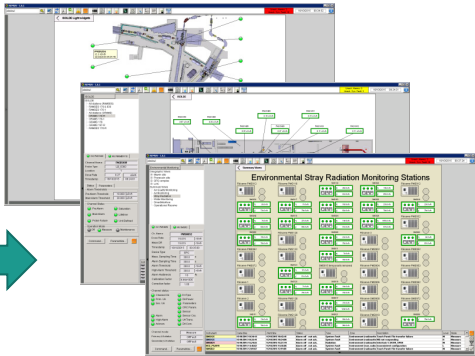
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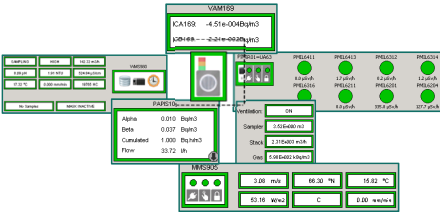
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Tailored User Interfaces



Users build their own synoptic views hierarchies

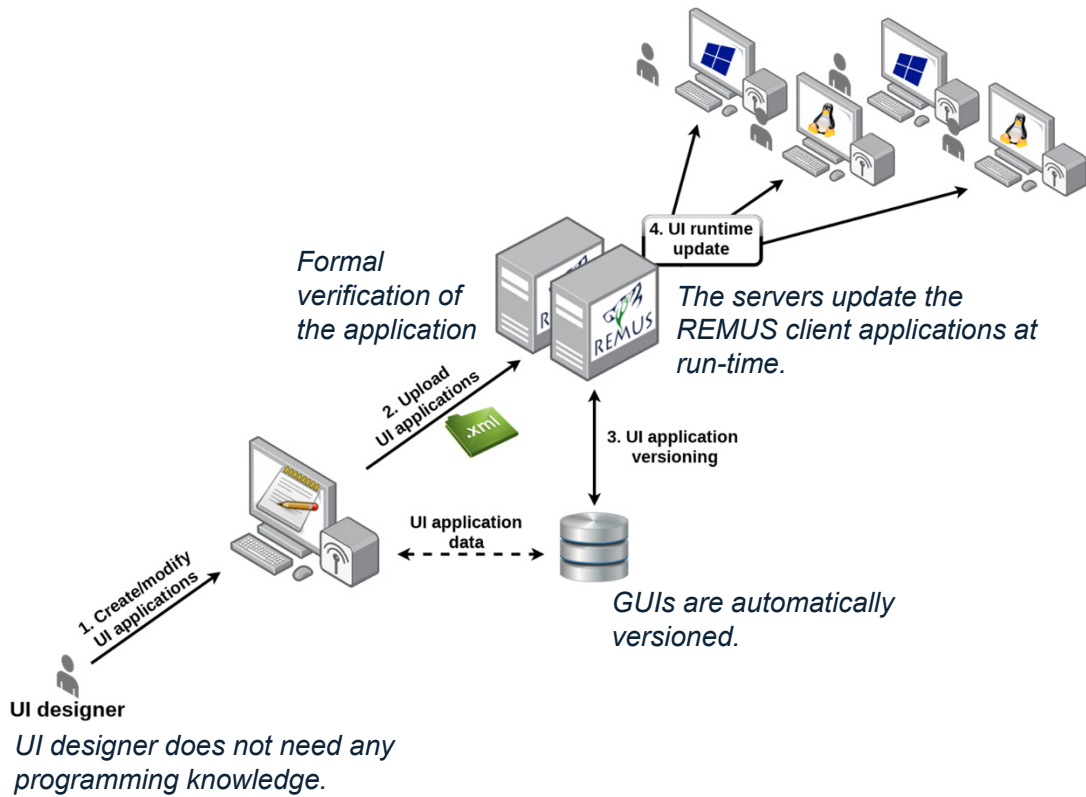


The GUI editor proposes a catalogue of widgets to drag and drop into synoptic views

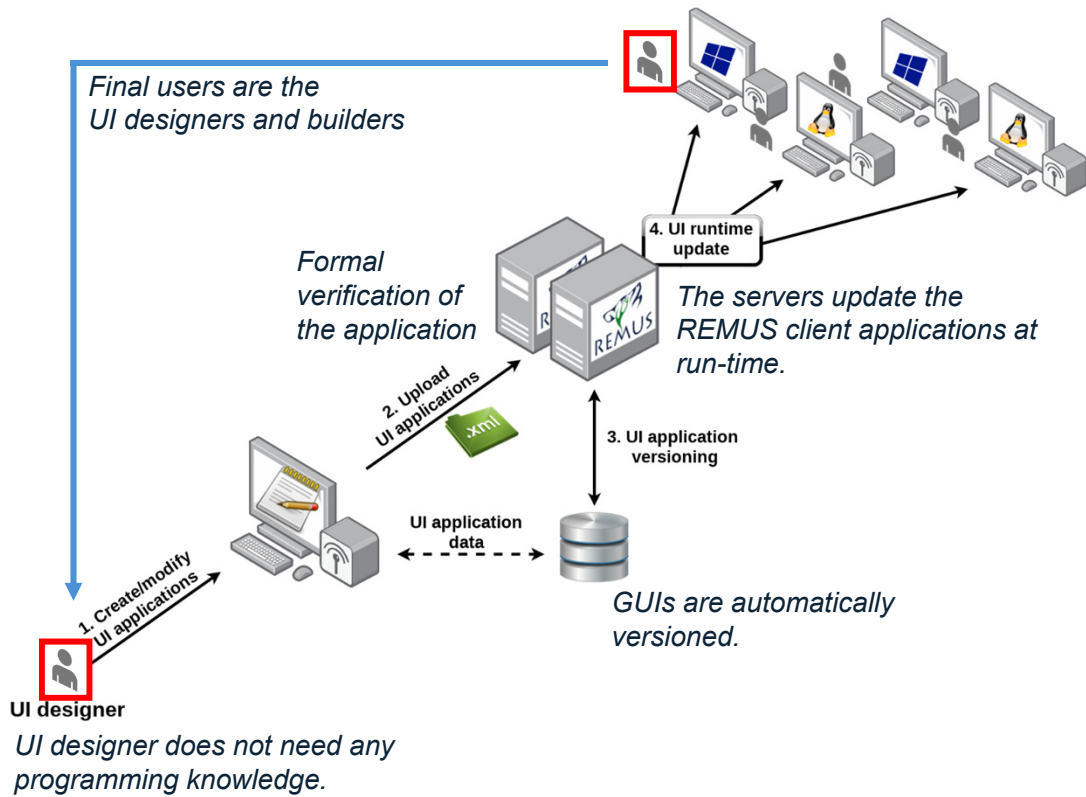


Access control and user roles (visualization, parametrization, administration, etc.)

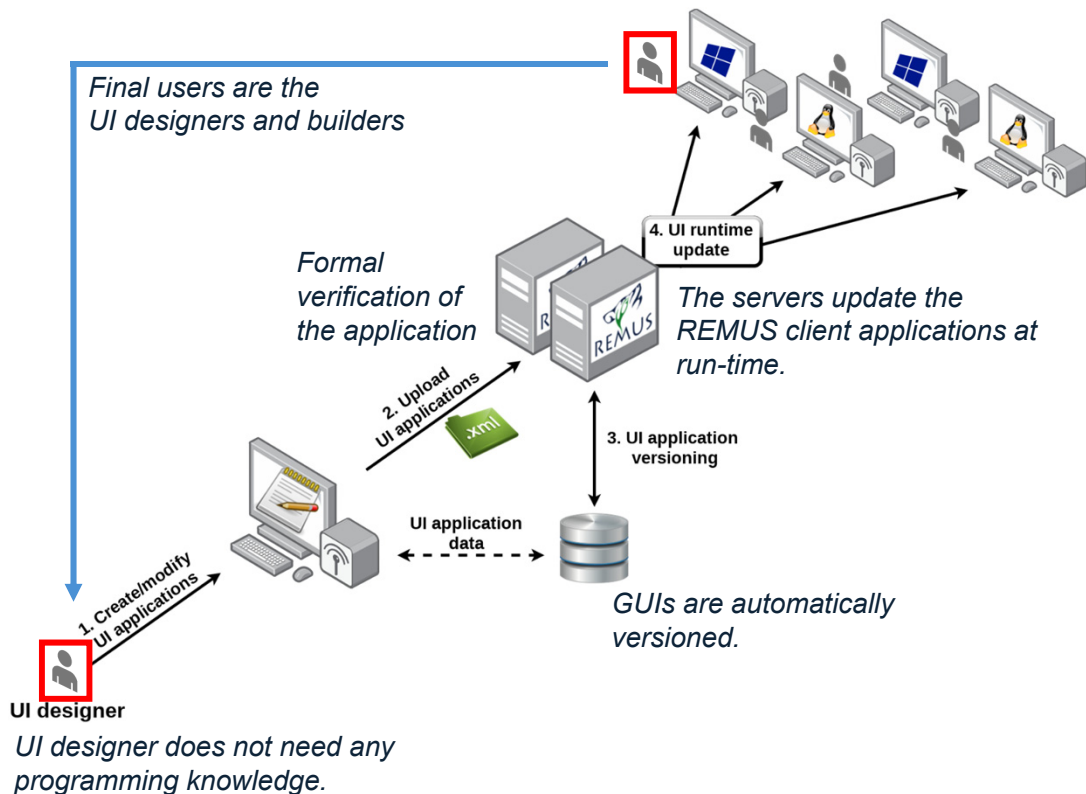
REMUS - Applications Run-Time modification and upload



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- Creating or modifying a REMUS application is just using a function of the system.
- Reduced effort in testing.
- The Application Editor software ensures that the application is valid and well-formed.
- The REMUS servers parse the UI (.xml) to:
 - Validate that the new UI is coherent and well-formed.
 - Construct the UI in run-time

Appropriate for Safety Systems

REMUS - Continuous Operation

■ Challenges

- HSE needs to operate its supervisory system 24/7, 365 days a year:
 - Monitoring is necessary during accelerator runs AND shutdowns



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■ Solution implemented in REMUS

- Redundancy of the Servers and all sub-systems
- All software components are designed to be resilient
- Maintenance and support operations designed to be executable at Run-Time:
 - Deployment of new User Interfaces
 - Declaration and connection of new equipment in the system (“plug-and-play”)
 - Framework upgrades
 - Operating System upgrades



REMUS - Continuous Operation

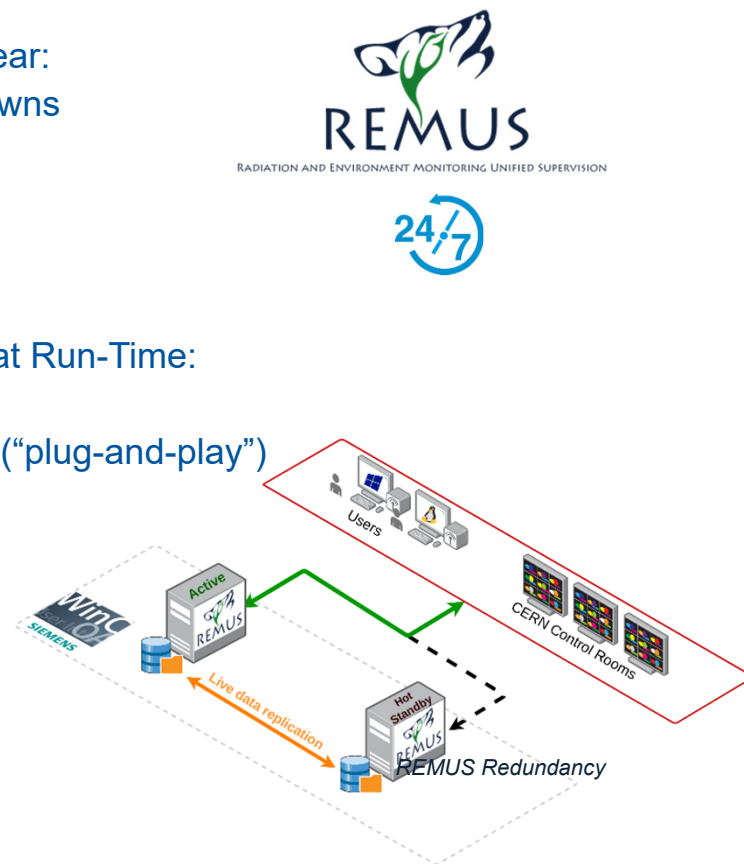
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REMUS availability 2013-2018: >99.998%

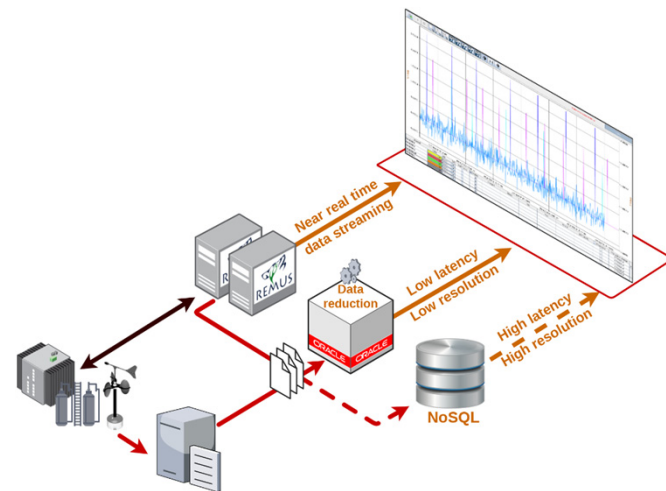


REMUS - Future evolutions

- **More equipment to interface!**

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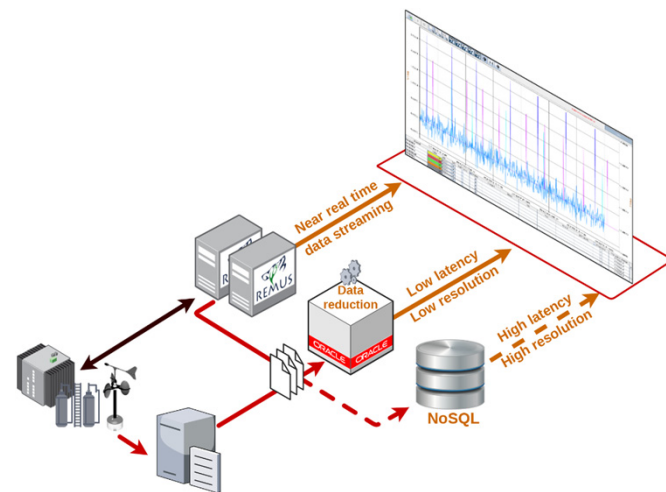
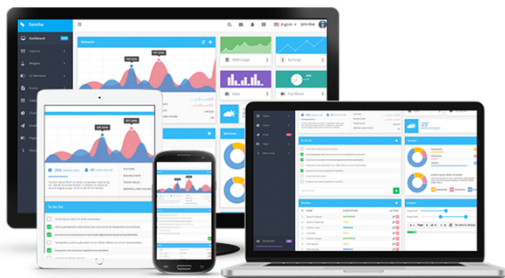
- **More equipment to interface!**
- **Big Data & modern data Streaming Services**
 - Allows performant Real Time data streaming service
 - Allows high resolution measurement archiving, without data reduction



REMUS future Data service architecture

REMUS - Future evolutions

- **More equipment to interface!**
- **Big Data & modern data Streaming Services**
 - Allows performant Real Time data streaming service
 - Allows high resolution measurement archiving, without data reduction
- **Web & Mobile Applications (read-only)**
 - Required by users for on-site interventions



REMUS future Data service architecture

Thank you!